

Observation Theory

Geometry, Distortion, and Reliability as Properties of the Observer

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Abstract

We unify and operationalize a family of results under one thesis—*Observation Theory*: geometry, distortion, and reliability are properties of the *observation* (the consumer’s read operator, budget, and channel), not of the object observed. From a second-order expansion of the downstream metric we recover the single controlling quantity, the reconstruction error read through the consumer, $d_{\mathcal{O}} = \text{tr}(P_{\mathcal{C}}\Sigma_{\delta})$, with Σ_{δ} the error covariance and $P_{\mathcal{C}} = \mathbb{E}[J^{\top}GJ]$ the Fisher-metric-weighted sensitivity. This weighted-quadratic fidelity is classical: it is the indirect/remote source-coding distortion of Dobrushin–Tsybakov and Wolf–Ziv, reduces to Shannon’s Gaussian rate–distortion by Berger’s change of variables, and is Hessian-weighted loss-aware compression (OBD/OBS through GPTQ) written as a metric. Our contribution is the *middle term*: $P_{\mathcal{C}}$ is a geometric, PSD, water-fillable operator that is *identifiable from the consumer alone*, sitting between Shannon rate–distortion ($P_{\mathcal{C}} = I$) and the information bottleneck. We make that middle term testable and test it, under sealed, before-run registrations with ex-ante bars: distortion— $\text{tr}(P_{\mathcal{C}}\Sigma_{\delta})$ governs downstream error, demonstrated and replicated across softmax attention, retrieval ranking, and gradient optimization, with a reconstruction-identical code *flipping* its verdict when the consumer changes; identifiability—blind planted-subspace recovery at $16\times$ chance, validated against ground truth; capacity—an extreme-value vacuity threshold that locates retrieval death to a few percent across 14 corpora; and rate—a fixed-budget verdict that *inverts* under budget-matched observation on real embedding manifolds. The analytic consumers are *synthetic* by design (planted read operators, chosen query banks, constructed Hessians), including an $H=I$ control where reconstruction becomes exactly optimal; the one real-substrate face is the budget inversion, and hardening on trained-model consumers is the decisive open validation. A fifth conjecture—a density quotient restoring the invariant in non-spectral retrieval—is *refuted* across four prospective tests and carried as a boundary. We prove achievability and converse of the consumer-relative rate–distortion function for a general source (Appendix A), and for discrete sources exhibit a consumer-lossless floor below the source entropy. Shannon is the corner where the observer reads everything.

1 Introduction

Shannon fixed a distortion measure and asked for the fewest bits meeting it [1, 2]; Gauss fixed the same measure and asked for the least-squares estimate [3]. This paper reads their common assumption as the variable of interest: the error that matters is not the one a code makes but the one a consumer *reads*, and different consumers read different subspaces of the same error. Once the distortion is the task’s, reconstruction is a single corner.

The idea that one should code a source to serve a *functional* of it, not the source, is not new: it is the indirect (remote) rate–distortion problem [5, 6, 7]; its lossless, discrete form is coding

for computing and graph entropy [8, 9]; its modern face is the rate–distortion–perception tradeoff [10] and task-oriented / semantic communication [11]; and its deployed form, for neural networks, is loss-aware (Hessian-weighted) compression, from Optimal Brain Damage/Surgeon [16, 17] to Hessian-aware quantization and GPTQ [18, 19]. What is missing across these is a single, *identifiable geometric* object that names the task’s metric and lets one predict, recover, and test it. Observation Theory supplies it: the read operator $P_{\mathcal{C}}$, the local Fisher metric of the consumer, sitting as the geometric middle term between Shannon rate–distortion ($P_{\mathcal{C}} = I$) and the information bottleneck [12] (an arbitrary, non-geometric relevance variable). We contribute (i) the map placing these literatures on one axis; (ii) $P_{\mathcal{C}}$ ’s *identifiability*—recovery from the consumer alone, validated blind against a planted ground truth, which the quantization literature (estimating the Hessian from calibration data) never validates against truth; (iii) an ex-ante extreme-value reliability certificate; (iv) budget-relativity of the verdict; and (v) a registration-first empirical program—sealed predictions, ex-ante bars, a refuted face carried as prominently as the wins—whose recurring demonstration is a single code whose downstream verdict *inverts* when nothing changes but the consumer. The three companion works are Paper I (the angular observer and the spectral budget-inversion mechanism) [20], Paper II (geometric key quantization and the reconstruction-is-not-the-target negative) [21], and the claim ledger [22]; this is their information-theoretic synthesis.

2 The controlling quantity

Definition 1 (Instance). *An instance is $(X, \mathcal{C}, \mathcal{Q})$: a representation with vectors $x \in \mathbb{R}^d$, a consumer $\mathcal{C} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ with a divergence D_Y on its outputs, and a family $\mathcal{Q}$ of lossy codes producing $\hat{x} = x + \delta$. The downstream distortion is $d_{\mathcal{O}} := D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x}))$, measured on the consumer output, never assumed from the ambient metric of X .*

Proposition 1 (Consumer-projected distortion). *Let D_Y be twice differentiable with $D_Y(u, u) = 0$, $\nabla_v D_Y(u, v)|_{v=u} = 0$, and information metric $G(u) = \nabla_v^2 D_Y(u, v)|_{v=u} \succeq 0$; let $J(x) = \partial \mathcal{C} / \partial x$. For error $\delta = \hat{x} - x$ with conditional covariance $\Sigma_{\delta}(x)$, the exact leading-order distortion is*

$$d_{\mathcal{O}} = \frac{1}{2} \mathbb{E}_x \text{tr}(P_{\mathcal{C}}(x) \Sigma_{\delta}(x)) + o(\|\delta\|^2), \quad P_{\mathcal{C}}(x) = J(x)^{\top} G(\mathcal{C}(x)) J(x) \succeq 0.$$

When $\Sigma_{\delta}(x)$ is uncorrelated with the point-dependent weight $P_{\mathcal{C}}(x)$ this factors as $\frac{1}{2} \text{tr}(P_{\mathcal{C}} \Sigma_{\delta})$ with $P_{\mathcal{C}} = \mathbb{E}_x[P_{\mathcal{C}}(x)]$, $\Sigma_{\delta} = \mathbb{E}_x[\Sigma_{\delta}(x)]$; reconstruction distortion $\mathbb{E}\|\delta\|^2 = \text{tr}(\Sigma_{\delta})$ is the case $P_{\mathcal{C}} = I$.

Proof. Expand $\mathcal{C}(x + \delta) = \mathcal{C}(x) + J\delta + O(\|\delta\|^2)$ and D_Y to second order about its minimum; the value and gradient terms vanish, leaving $\frac{1}{2} \delta^{\top} (J^{\top} G J) \delta + o(\|\delta\|^2)$. Take $\mathbb{E}_x$; the factored form requires the stated uncorrelatedness (for real quantizers both J and Σ_{δ} depend on x , so the exact object is the x -averaged trace). $\square$

We keep the exact form $\mathbb{E}_x \text{tr}(P_{\mathcal{C}}(x) \Sigma_{\delta}(x))$ as primary and use the factored $\text{tr}(P_{\mathcal{C}} \Sigma_{\delta})$ as the convenience it is. The theory is this one line: the distortion governing a task is the error covariance read through the consumer’s sensitivity. The rest derives $P_{\mathcal{C}}$, shows it recoverable, and studies the observer’s channel and budget. (We drop the harmless $\frac{1}{2}$.)

3 The read operator, derived

Proposition 2 (Softmax attention). *Read keys $\{k_s\}_{s=1}^S$ through a query bank: logits $l_s = q^{\top} k_s / \sqrt{d}$, output $p = \text{softmax}(l)$, $D_Y = \text{KL}$. Under key perturbation δ_s ,*

$$d_{\mathcal{O}} = \frac{1}{2d} \mathbb{E}_q \text{Var}_s(q^{\top} \delta_s) + O(\|\delta\|^3) = \frac{1}{2d} \text{tr}(Q \bar{\Sigma}_{\delta}) + O(\|\delta\|^3),$$

$Q = \mathbb{E}_q[qq^\top]$ the query second moment, $\bar{\Sigma}_\delta = \mathbb{E}_q \sum_s p_s \tilde{\delta}_s \tilde{\delta}_s^\top$ the p -weighted covariance of the across-token-demeaned errors $\tilde{\delta}_s = \delta_s - \sum_{s'} p_{s'} \delta_{s'}$.

Proof. The Fisher metric of softmax is $G = \text{diag}(p) - pp^\top$, so $\text{KL}(p \parallel \text{softmax}(l + \Delta l)) = \frac{1}{2} \text{Var}_s^p(\Delta l_s) + O(\Delta l^3)$ with $\Delta l_s = q^\top \delta_s / \sqrt{d}$. The $-pp^\top$ term is the softmax's invariance to a common logit shift; it removes the p -mean, giving $\tilde{\delta}_s$. Since p depends on q , the outer $\bar{\Sigma}_\delta$ carries the $\mathbb{E}_q$ and the final Q -factorization inherits the uncorrelatedness proviso of Prop. 1. $\square$

Two consequences are the phenomena we later measure: $P_C \propto Q$, so an isotropic bank gives $P_C \propto I$ (reconstruction-like) while a subspace bank projects—the read operator, hence the winning code, changes with the query distribution; and only the *demeaned* error survives, so the across-token mean is a nuisance the softmax cannot see. This last fact resolves an apparent tension with the DC-offset collapse of Paper II [21]: a truly common error component is free by shift invariance, but a per-channel DC *offset* is not a common component—it inflates the quantizer's range, wasting levels and enlarging the *visible* per-token error, which is exactly why an asymmetric zero-point (asym-NF4) repairs it at no bit cost. Invisible-to-the-consumer and free-to-encode are the same statement; range waste is a different one.

Proposition 3 (Gradient-descent optimizer; exact). *Let $L(\theta) = \frac{1}{2} \theta^\top H \theta$, $g = H\theta$, and $\hat{g} = g + \delta$ used in a step of size η . Then $L(\theta - \eta \hat{g}) - L(\theta - \eta g) = \frac{1}{2} \eta^2 \delta^\top H \delta - \eta \delta^\top H(\theta - \eta g)$, whose error-quadratic part is $\frac{1}{2} \eta^2 \text{tr}(H \delta \delta^\top)$; the linear cross term vanishes in expectation over zero-mean error, giving $\mathbb{E}[d_{\mathcal{O}}] = \frac{1}{2} \eta^2 \text{tr}(H \Sigma_\delta)$, so $P_C = H$ exactly.*

This is loss-aware pruning/quantization made a metric: Hessian-weighted deletion [16, 17, 18, 19] is $\text{tr}(H \Sigma_\delta)$, and calibration-based Hessian estimation is the production form of the probe of §5. Reconstruction is the flat case $H = I$.

Remark (Retrieval). For score-ranking $s(q, x_i) = q^\top x_i$, the inverted-pair fraction is, to leading order for smooth rank statistics, monotone in $\text{tr}(P_C \bar{\Sigma}_\delta) / \text{tr}(P_C \bar{\Sigma}_0)$, with $P_C = \mathbb{E}[qq^\top]$, $\bar{\Sigma}_\delta = \text{Cov}_i(\delta_i)$ the across-item error covariance and $\bar{\Sigma}_0 = \text{Cov}_i(x_i)$ the signal covariance (both demeaned across items); P_C is a subspace projector when queries concentrate. Rank statistics are non-smooth; we treat this instance empirically (§4).

4 Distortion: the consumer is the metric (GO-2)

Proposition 2 predicts that at matched reconstruction the ranking of two codes is set by $\text{tr}(P_C \Sigma_\delta)$, hence by the query distribution.

Dissociation. Two 4-bit codes at an audited matched budget (spread 0.19 bit) with *identical* reconstruction (0.0938 vs 0.0934): the whole-vector code is 2.53× worse on softmax-KL (12/12 seeds); a wrong-quotient control is 21× worse at $\sim 5\times$ the reconstruction. Reconstruction is not the controlling quantity ([demonstrated]).

Flip. Holding reconstruction fixed and changing only the query bank, the ranking *inverts*: isotropic $\Rightarrow$ code *a* better (2.5×, 12/12); subspace $\Rightarrow$ code *b* better (12/12). The directly computed $\text{tr}(P_C \Sigma_\delta)$ tracks the sign on every seed in both consumers; reconstruction, a consumer-blind scalar, cannot (rank correlation 0.40 vs 1.00). This is Proposition 2 with Q swapped ([demonstrated]).

Replication and out-of-sample. The flip replicates on retrieval ranking (different representation, set-valued consumer): identical reconstruction (0.0964), ranking flips with the read subspace (log-advantage -4.70 vs $+4.65$, 12/12), $\text{tr}(P_C \Sigma_\delta)$ tracking, reconstruction blind ([**replicated**]). It generalizes to optimization with $P_C = H$ (Prop. 3): at matched reconstruction the curvature-metric distortion flips between two Hessians (12/12), tracked by $\text{tr}(H \Sigma_\delta)$, reconstruction uninformative (-0.20)—and under $H = I$ reconstruction tracks *perfectly* (1.00), the derived $P_C = I$ corner observed ([**predicted**], out-of-sample). One derivation; three read operators. These consumers are *synthetic*—the point is to validate the theory’s logic (the $H = I$ positive control being the clean confirmation); the trained-model hardening (§11) is the outstanding validation.

5 Identifiability: the metric is recoverable (GO-1)

Proposition 4 (Recovery of the read subspace). *Let $\mathcal{C}(x) = f(Wx)$, $W \in \mathbb{R}^{r \times d}$ of rank r , f coordinatewise with $\mathbb{E}[f'_i(Wx)^2] > 0$ for every coordinate i (no saturated unit). Then $\mathbb{E}_x[J^\top J] = W^\top \mathbb{E}[\text{diag}(f'^2)]W$ has rank r with column space $\text{row}(W)$, so its top- r eigenvectors span the read subspace; a finite-difference estimate $\hat{J}u \approx (\mathcal{C}(x+hu) - \mathcal{C}(x-hu))/2h$ recovers it from consumer queries alone.*

We built such a probe—receiving only the callable $\mathcal{C}(\cdot)$ and the dimensions, never W , the codes, or d_O (blinding enforced in code). It recovers the planted read subspace at overlap 0.936 against a 0.059 chance baseline ($\approx 16\times$), and the *blind* $\text{tr}(\hat{P}_C \Sigma_\delta)$ predicts the flip of §4 on 12/12 seeds in both consumers with rank correlation 1.00—identical to ground truth—while reconstruction cannot (0.40) ([**predicted**]). Loss-aware compression estimates such a metric from calibration data; the contribution here is validating *recovery against a planted ground truth under enforced blinding*, and composing it: from the consumer alone one predicts which code wins, before compressing anything.

6 Capacity: a vacuity threshold, derived (GO-3)

Proposition 5 (Retrieval vacuity threshold). *Model the $N-1$ distractor scores, standardized to mean 0 and variance 1, as i.i.d. standard Gaussian (a reference; dependence or heavier tails shift the constants). Top-1 success needs the standardized true margin $\hat{\mu}$ to exceed the distractor maximum M_{N-1} , so $\text{recall@1} = \Pr[\hat{\mu} > M_{N-1}]$ crosses $\frac{1}{2}$ at the median of the maximum,*

$$\mu_{\text{crit}} := \text{med}(M_{N-1}) = a_N - b_N \ln \ln 2, \quad a_N = \sqrt{2 \ln(N-1)}, \quad b_N = 1/a_N,$$

with $\hat{\mu} = (\overline{\text{true}} - \overline{\text{distr}})/\text{std}$ the margin certificate and $\rho = \hat{\mu}/\mu_{\text{crit}}$; the certificate is vacuous for $\rho < 1$.

Derivation. By Fisher–Tippett, standardized i.i.d. Gaussian maxima converge to a Gumbel law of location a_N , scale $b_N = 1/a_N$. For a fixed margin, $\text{recall@1} = F_{\max}(\hat{\mu})$ equals $\frac{1}{2}$ at the median $a_N - b_N \ln \ln 2$. The mean $\mathbb{E}[M_{N-1}] = a_N + \gamma b_N$ (γ Euler–Mascheroni) differs from the median at order $b_N = \Theta(1/\sqrt{\ln N})$; the $\frac{1}{2}$ -crossing is the median, not the mean. $\square$

Across 14 structurally distinct corpora this *derived* (not fitted) Gaussian-reference threshold locates death: the recall- $\frac{1}{2}$ crossing sits at $\rho = 0.948$, slightly *below* 1—exactly as score dependence predicts, since correlated or heavier-tailed distractors reduce the effective number of independent distractors and lower the true median below the i.i.d.-Gaussian reference. The certificate orders corpora by recall at rank correlation 0.991, beating the un-normalized margin (0.873), with a finite-width transition ([**demonstrated**]). Capacity says, from a computable quantity, *where* the observation channel dies before any retrieval is run.

7 Rate: the verdict belongs to the budget (GO-4)

The read operator, capacity, and distortion are stated at a fixed observer budget; the budget is a rate, and it inverts the verdict. The mechanism is spectral: an eigenfunction φ_k with eigenvalue λ_k oscillates on the wavelength $\lambda_k^{-1/2}$, and Weyl’s law $N(\lambda) \sim c \text{Vol}(M) \lambda^{d_M/2}$ (d_M the intrinsic dimension) grows the count of modes below a fixed scale with resolution.

Lemma 1 (Fixed-band consistency). *At a fixed low-mode budget m , as the sample grows the graph operator converges spectrally to the manifold operator [14], so the fixed low- λ band sharpens and geodesic fidelity is nondecreasing (the mechanism proved in Paper I, [20]).*

Holding instead the resolved scale τ fixed (spectral-gap matching), $m = N(\tau)$ grows with N ; the newly resolved modes near τ have wavelength approaching the sampling scale and, for a non-ideal substrate, inject sub-geodesic detail, so geodesic fidelity is nonincreasing—Paper I’s registered inversion [20]. We replicate it on a new substrate: 768-dimensional multilingual book-paragraph embeddings (Project Gutenberg translations; 32,000 points, three seeds). Fixed $m = 10$: geometry preservation rises with N (trend +0.4, +1.0, +1.0). Spectral-gap-matched, $m : 10 \rightarrow 121/126/159$: geometry *collapses* (trend -1.0 every seed); the budgets agree at the smallest N (gap ≤ 0.002) ([**replicated**]). Opposite verdicts about the same data, set by the observer’s rate: “clean geometry” is a relation between observer resolution and substrate scale, not a substrate invariant.

8 A consumer-relative rate–distortion function

The consumer-projected distortion is the single-letter measure $d_C(x, \hat{x}) = (x - \hat{x})^\top P_C(x - \hat{x})$; define $R_C(D) = \inf\{I(X; \hat{X}) : \mathbb{E} d_C(X, \hat{X}) \leq D\}$. This is a weighted-quadratic (difference) distortion, and its theory is classical [4].

Proposition 6 (Gaussian rate–distortion; Berger reduction). *For $X \sim \mathcal{N}(0, \Sigma_x)$, in the basis diagonalizing $P_C^{1/2} \Sigma_x P_C^{1/2}$ with eigenvalues γ_i ,*

$$R_C(D) = \sum_i \frac{1}{2} \log_2^+ \frac{\gamma_i}{\theta}, \quad \sum_i \min(\gamma_i, \theta) = D,$$

a reverse water-filling on the P_C -weighted source spectrum. For $P_C = I$ this is Shannon’s Gaussian $R(D)$ [2]; the estimation dual is Gauss–Markov least squares [3].

Proof. $\text{tr}(P_C \Sigma_\delta) = \|P_C^{1/2}(x - \hat{x})\|^2$; with $y = P_C^{1/2}x$ this is MSE on $y \sim \mathcal{N}(0, P_C^{1/2} \Sigma_x P_C^{1/2})$, and Shannon water-filling applies. The transform is invertible off $\ker P_C$, on which the consumer is blind and no rate is spent. $\square$

Bits water-fill on directions the consumer reads *and* the source varies, spending nothing on $\ker P_C$ —why a reconstruction-optimal code ($P_C = I$ allocation) is beaten by a consumer-matched one when $P_C \neq I$, and why the winner *flips* with P_C (§4). The $H = I$ control confirms it: only at $P_C = I$ is reconstruction the right objective, and there it is exactly right.

Proposition 7 (Consumer-lossless floor; discrete source). *Let X be discrete with finite entropy, Π the projector onto $\text{range}(P_C)$. There exist codes with $\text{tr}(P_C \Sigma_\delta) = 0$ at rate*

$$R_C(0) = H(\Pi X) \leq H(X), \quad H(X) - R_C(0) = H(X | \Pi X) \geq 0,$$

with equality iff Π is injective on $\text{supp}(X)$; the saving is strict exactly when distinct symbols differ only within the consumer-invisible subspace $\ker P_C$.

Proof. $\text{tr}(P_C \Sigma_\delta)$ depends on δ only through $\Pi\delta$; sending ΠX losslessly and reconstructing $(I - \Pi)X$ by its conditional mode gives $\Pi\delta = 0$, zero distortion, at rate $H(\Pi X)$. If Π collapses two support points, $H(\Pi X) < H(X)$. $\square$

Remark (Continuous sources). For a continuous source lossless coding costs infinite rate ($R_C(D) \rightarrow \infty$ as $D \rightarrow 0$, consistent with Prop. 6). The surviving statement is that at *every* $D > 0$ the water-filling of Prop. 6, equivalently the reduction of Cor. 1, spends zero rate on $\ker P_C$: the operational rate depends only on the r -dimensional tilted source. This lossless-to-the-consumer phenomenon is exactly coding for computing and graph entropy in the discrete world [8, 9], here given a geometric read operator.

Placed on one axis: Shannon rate–distortion ($P_C = I$) $\subset$ consumer-relative rate–distortion (P_C a rank- r geometric, *identifiable* operator, Props. 6,7) $\subset$ the information bottleneck [12]. The middle term is the operational one— P_C is PSD, water-fillable, and recoverable from the consumer (Prop. 4)—and its coding theorem (achievability and converse, general source) is Appendix A.

9 A boundary: the refuted quotient (GO-5)

Not every plausible face survives a sealed prediction. In the spectral setting the $\alpha=1$ anisotropic normalization of a diffusion operator cancels sampling density and recovers density-free geometry [13]: there the density mode lies in $\ker P_C$. We conjectured this density quotient likewise restores the invariant in a *non-spectral* domain (locally-scaled ADC retrieval, a hubness correction [15]). It does not, across four prospective tests (direct affinity twice, ADC on real embeddings, non-spectral diffusion-distance). The reason is Proposition 6: for a direct inner-product consumer the density is *read*—it is not in $\ker P_C$ —so quotienting it removes signal and manufactures anti-hubs; in the diffusion domain the direction is right ($\alpha=1$ optimal, recovered non-spectrally) but the effect is under 2% and *not density-specific* (a uniform-density control gains as much). Per its sealed registration the conjecture is [**refuted**]. The sharpened statement—the $\alpha=1$ cancellation is a property of the diffusion operator and does not transfer to direct observation—is Proposition 6 read backwards: rate is saved on $\ker P_C$ only when the nuisance actually lies there. The refutation sharpens the theory rather than embarrassing it.

10 Honest negatives and method

Eleven standing negatives bound the theory and are reported with the prominence of the positives. Reconstruction cosine is not a proxy for key quality (Paper II [21], the motivating negative). The flip is observable only for reconstruction-*matched* codes: absent a tie one code Pareto-dominates and reconstruction is not falsified (a prospective miss locating a precondition, consistent with the water-filling of Prop. 6). The projected-variance trace governs the discriminating comparison but is not a complete rank statistic over all codes. The density quotient (§9). Two are methodological confessions: twice a substantive claim passed at 12/12 while an over-strict secondary gate we had written failed, and twice a degenerate substrate was caught only by a regime-coverage gate.

Every empirical claim was governed by a dated, hashed registry entry committed—sealed—before the first measurement it governs; bars were fixed ex ante; misses enter the ledger as misses; amendments only dated and logged before unblinding; prospective claims (§5,7,9) were blinded where feasible. The house rule is that the paper may not assert what the ledger cannot show.

11 Discussion

Observation Theory asks a change of default. Where a pipeline minimizes reconstruction and hopes the task follows, the theory says: recover the consumer's read operator (§5), water-fill bits on P_C rather than on I (§8), respect the channel's vacuity threshold (§6), and remember the verdict is relative to the observation budget (§7). Against the literatures it unifies—indirect source coding, coding for computing, rate–distortion–perception, task-oriented communication, loss-aware compression—the contribution is making the task's metric a single identifiable geometric object and testing it adversarially. The decisive open validation is not conceptual but empirical: the analytic consumers of §§4–5 are synthetic, and the one run that converts “the logic is validated” into “the theory works where it matters” is the keys experiment on an actual trained attention layer, with the query second moment recovered by the blind probe and the winning quantizer predicted before compression—Gate B for the book. Open theory: the achievability and converse of $R_C(D)$ beyond the Gaussian case we settle in Appendix A; whether the flip is exactly its argmin migrating with P_C ; and the consumer-relative *channel* coding theorem behind §6. Four faces, one refutation, eleven negatives: a world to map, now with coordinates in the literature.

A The consumer-relative coding theorem

Lemma 2 (Reduction to the tilted source). *Let $Y = P_C^{1/2}X$, on $\text{range}(P_C)$ of dimension $r = \text{rank } P_C$. Then $d_C(x, \hat{x}) = \|y - \hat{y}\|^2$ with $\hat{y} = P_C^{1/2}\hat{x}$, and $R_C(D) = R_Y(D)$, the MSE rate–distortion function of Y ; $\ker P_C$ costs zero rate.*

Proof. $d_C = \|P_C^{1/2}(x - \hat{x})\|^2 = \|y - \hat{y}\|^2$. Objective and constraint depend on $\hat{X}$ only through $\hat{Y}$, so by data processing $I(X; \hat{X}) \geq I(Y; \hat{Y})$; the minimizer places all information in Y (a channel $p(\hat{y} | y)$, then $\hat{X} = P_C^{-1/2}\hat{Y}$ on the range, conditional mean on $\ker P_C$), attaining equality. Hence the minimum is $R_Y(D)$. $\square$

Theorem 1 (Achievability and converse). *Let $\{X_t\}$ be i.i.d. $\sim p_X$ with $\mathbb{E} d_C(X, \hat{x}_0) < \infty$ and $\mathbb{E} d_C(X, \hat{x}_0)^2 < \infty$ for some $\hat{x}_0$. For every $D > 0$: (i) for any $R > R_C(D)$ there are $(2^{nR}, n)$ codes with $\limsup_n \mathbb{E} \frac{1}{n} \sum_t d_C(X_t, \hat{X}_t) \leq D$; (ii) every code of expected distortion $\leq D$ has rate $\geq R_C(D)$; and R_C is convex and nonincreasing.*

Proof. Achievability. Fix $\varepsilon > 0$ and a test channel $p^*(\hat{x} | x)$ with $\mathbb{E} d_C \leq D$, $I(X; \hat{X}) \leq R_C(D) + \varepsilon$; marginal $p_{\hat{X}}$. Draw $\{\hat{X}^n(w)\}_{w=1}^{2^{nR}}$ i.i.d. $\sim \prod_t p_{\hat{X}}$. Let $A_\varepsilon^{(n)}$ be the distortion-typical set (empirical entropies within ε of $h(X), h(\hat{X}), h(X, \hat{X})$ and $\frac{1}{n} \sum_t d_C \leq \mathbb{E} d_C + \varepsilon$). Encode x^n by any w with $(x^n, \hat{X}^n(w)) \in A_\varepsilon^{(n)}$, else $w = 1$. By the covering lemma a random codeword is jointly typical with a typical x^n with probability $\geq 2^{-n(I+3\varepsilon)}$, so $\Pr[\text{no jointly typical codeword}] \leq (1 - 2^{-n(I+3\varepsilon)})^{2^{nR}} \leq \exp(-2^{n(R-I-3\varepsilon)}) \rightarrow 0$ for $R > I + 3\varepsilon$. On success the per-letter distortion is $\leq D + \varepsilon$; with failure probability $p_f \rightarrow 0$, Cauchy–Schwarz bounds the failure contribution $\mathbb{E}[\frac{1}{n} \sum_t d_C \mathbf{1}_{\text{fail}}] \leq \sqrt{\mathbb{E}[(\frac{1}{n} \sum_t d_C)^2]} \sqrt{p_f} \rightarrow 0$ (finite by the second-moment hypothesis). Thus $\limsup_n \mathbb{E} \frac{1}{n} \sum_t d_C \leq D + \varepsilon$; let $\varepsilon \rightarrow 0$.

Converse. With $D_t = \mathbb{E} d_C(X_t, \hat{X}_t)$, $nR \geq H(W) \geq I(X^n; \hat{X}^n) \geq \sum_t I(X_t; \hat{X}_t) \geq \sum_t R_C(D_t) \geq nR_C(\frac{1}{n} \sum_t D_t) \geq nR_C(D)$, by memorylessness, the definition of R_C , Jensen (convexity), and monotonicity. Convexity and monotonicity follow from time-sharing and distortion relaxation. $\square$

Corollary 1. $R_C(D)$ equals the MSE rate–distortion of the r -dimensional tilted source $Y = P_C^{1/2}X$; for Gaussian X it is Prop. 6; discretely, $R_C(0) = H(\Pi X)$ (Prop. 7). The operational rate to serve

a consumer is governed entirely by the source projected into the read subspace—the precise sense in which Shannon’s theorem is the corner $P_C = I$, where the read subspace is everything.

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Reproducibility & provenance. All results resolve to sealed, before-run registry entries and committed artifacts in the companion repository [22], private pending release; the release will be sealed to an external, immutable snapshot (Zenodo/OSF) so that registry hashes are anchored independently of forgeable commit timestamps and the custody chain is protected against history rewriting. Computational experimentation and drafting were carried out with an AI research assistant under the registration-first protocol of §10.